CORE PURE / VECTORS, MATRICES & LINEAR TRANSFORMATIONS

Invariant Points and Lines

Mark scheme 9 questions · 75 marks

Questions

Question	Marks by part	Marks	Page
Question 1	(a) 2 (b) 2	4	4
Question 2		5	5
Question 3	(a) 4 (b) 5	9	6
Question 4	(a) 2 (b) 2 (c) 3 (d) 2	9	7
Question 5	(a) 4 (b) 2 (c) 7	13	8
Question 6	(a) 5 (b) 3	8	9
Question 7	(a) 3 (b) 4	7	10
Question 8	(a) 5 (b) 2	7	11
Question 9	(i) 8 (ii) 2 (iii) 3	13	12
	Total	**75**	

Reading the mark scheme

These rules apply throughout, unless a note says otherwise.

Marks

Every mark is worth one, so **M1 A1** on a row is two marks. Only marks coded ft follow through.

Code	Meaning
M1	Method: a correct method, applied to this question. A slip doesn't cost it if the method is clear. A formula must be used, not just quoted.
dM1	Dependent method: as M1, but only if the method mark before it was earned, unless its note names another.
A1	Accuracy: a correct answer or step. It needs the method mark it follows.
A1ft	As A1, following through from an earlier answer that the note names.
A1 AG	The last mark of a part whose answer is printed in the question: complete working, with no errors.
B1	A correct value, fact or statement. It needs no method mark.
B1ft	As B1, following through from an earlier answer that the note names.
E1	A written explanation, interpretation or conclusion. Concluding a proof by induction needs every earlier mark.

Abbreviations

ft and AG qualify a code in the Marks column. The others appear in the notes.

ft	follow through	**AG**	answer given in the question
cao	correct answer only, with no follow-through	**cso**	correct solution only: no errors in the part
oe	or an equivalent form	**awrt**	any answer that rounds to this value
soi	seen, or implied by later working	**isw**	ignore working after a correct answer
SC	special case: credit outside the main method	**their**	the student's own earlier value
condone	a slip that doesn't cost the mark	[ ]	working the mark doesn't need
s.f.	significant figures	**d.p.**	decimal places

Reading the scheme

- Each row is a line of working, with the marks it earns beside it. Where a row carries two marks, each note starts with its code.
- **[*n*]** ends each part with its total.
- A shaded *Alternative* is a second method for the same total. Mark an answer by the method it follows.
- Any other valid method earns the equivalent marks, unless the question names a method or says *hence*.
- A sketch earns one B1 for each feature named below its figure.

Accuracy and answers

- Unless a question says otherwise, give answers to 3 significant figures, and angles in degrees to 1 decimal place. Angles in radians take 3 significant figures.
- A final answer with more figures is accepted if it rounds to the scheme's value; one with fewer is not. An answer spoiled by rounding too early loses its accuracy mark.
- Where a question asks for an exact answer, a decimal loses the final accuracy mark. Otherwise either is accepted.
- Wrong or missing units cost nothing unless the question asks for them.
- An omitted $\mathrm{d}\theta$, $\mathrm{d}x$ or $\mathrm{d}u$ in an integral is not penalised where the variable of integration is clear from the integrand and the limits. It is penalised where it is genuinely ambiguous: a substitution part-way through, two variables in play, or limits in the wrong variable.
- A correct answer with no working earns full marks, unless the question needs the working: show that, prove, hence, a method it names, or an exact answer. A wrong answer with no working earns nothing. Calculator answers are treated the same way.
- A correct answer, once reached, isn't spoiled by later working (isw).

Value of g

- Take $g = 9.8\,\mathrm{m\,s^{-2}}$ unless a question states a value, which is then the only one accepted. Answers that use $g = 9.8$ may be given to 2 or 3 significant figures.
- Where a question doesn't state g, an answer worked with $g = 10$ throughout earns the same marks, keeping any value printed in the question. The worked solutions list the final answers with $g = 10$.

Question	Scheme	Marks	Notes
1(a)	$p=-1$: $2(3)-2(4)=\ldots$, $1(3)+2(4)=\ldots$	**M1**	Substitutes $p=-1$ into $\mathbf{M}$ and multiplies by the column vector $(3,4)$. Condone one slip in an entry.
	$6-8=-2$, $3+8=11 \Rightarrow (-2,11)$	**A1 AG**	Correct matrix and both products leading without error to $(-2,11)$.
		[2]	
1(b)	$p=2$: $5x-2y=x$, $4x-y=y$	**M1**	Substitutes $p=2$ and sets $\mathbf{M}\binom{x}{y}=\binom{x}{y}$, forming at least one equation in x and y. Condone one slip in an entry.
	$4x-2y=0 \Rightarrow y=2x$, gradient 2	**A1**	
		[2]	
	Total for Question 1	**4**	

Question	Scheme	Marks	Notes
2	$(x,\ mx+c) \mapsto \big((1+4m)x+4c,\ (2+3m)x+3c\big)$ $(2+3m)x+3c = m\big((1+4m)x+4c\big)+c$	**M1**	Substitutes the image of a general point of $y=mx+c$ into $y=mx+c$. Condone slips in the matrix product.
	$2+3m = m+4m^2,\ \ 3c=(4m+1)c$	**M1 A1**	**M1** Equates coefficients of x and constant terms in their equation. **A1** Both equations correct, oe.
	$2m^2-m-1=0 \Rightarrow m=\ldots$	**M1**	Solves their three-term quadratic in m by factorising, formula, completing the square or calculator.
	$m=-\frac{1}{2},\ c=0 \Rightarrow y=-\frac{1}{2}x$	**A1**	$y=-\frac{1}{2}x$ oe, with $c=0$ found from the constant equation.
	Total for Question 2	**5**	

Question	Scheme	Marks	Notes
3(a)	$(p+2)(p-1)-(-5)(-2)=0$	**M1**	Forms $ad-bc$ for **M** in terms of p and sets it equal to zero. Condone one sign slip.
	$p^2+p-12=0$	**A1**	Correct three-term quadratic oe.
	$(p+4)(p-3)=0 \Rightarrow p=\ldots$	**dM1**	Solves their three-term quadratic by factorising with constants multiplying to their constant term, the formula or completing the square; calculator allowed.
	$p=3$ or $p=-4$	**A1**	Both values, and no others.
		[4]	
3(b)	$\begin{pmatrix}5 & -5\\ -2 & 2\end{pmatrix}\begin{pmatrix}x\\ mx\end{pmatrix}=\begin{pmatrix}5x-5mx\\ -2x+2mx\end{pmatrix}$	**M1**	Applies **M** to a general point of $y=mx$ and requires the image to satisfy $y=mx$. Condone slips in the product.
	$-2x+2mx=m(5x-5mx)$	**A1**	Correct equation oe; x may already be cancelled.
	$5m^2-3m-2=0 \Rightarrow m=\ldots$	**dM1**	Rearranges, after cancelling x, to a three-term quadratic in m and solves it by factorising, the formula or completing the square; calculator allowed.
	$y=x$	**A1**	An equation of the line; $m=1$ alone is not enough.
	$y=-\frac{2}{5}x$	**A1**	oe, such as $2x+5y=0$; $m=-\frac{2}{5}$ alone is not enough.
		[5]	
Total for Question 3		**9**	

Question	Scheme	Marks	Notes
4(a)	A reflection leaves every point on its mirror line unchanged	**E1**	States that points on the mirror line are unchanged by the reflection. Not for asserting only that a reflection has invariant points.
	The line of invariant points is the mirror line	**E1**	States that it is the mirror line; accept axis or line of reflection.
		[2]	
4(b)	$\frac{1}{5}(-3x-4y)=x, \quad \frac{1}{5}(-4x+3y)=y$ $\Rightarrow 2x+y=0$	**M1**	Equates the image of (x,y) under **A** to (x,y) and forms at least one linear equation in x and y. Condone sign slips.
	$y=-2x$	**A1**	Accept $y=-2x+0$. A0 for $2x+y=0$ alone.
		[2]	
4(c)	$P'=\frac{1}{5}(-12-28,\ -16+21)\ [=(-8,1)]$	**M1**	Multiplies **A** by the position vector of P. Condone one arithmetic slip.
	$\left(\frac{4+\text{“}-8\text{”}}{2},\ \frac{7+\text{“}1\text{”}}{2}\right)$	**dM1**	Finds the midpoint of P and their image.
	$(-2,4)$	**A1**	
		[3]	
4(d)	Perpendicular to the mirror line: $(-2)\left(-\frac{2}{b}\right)=-1 \Rightarrow b=\ldots$	**M1**	Sets the product of the gradient of the line and their mirror-line gradient equal to -1, oe. Condone a sign slip in $-\frac{2}{b}$.
	$b=-4$	**A1**	
		[2]	
	Total for Question 4	**9**	

Question	Scheme	Marks	Notes
5(a)	Reflection in $y = \frac{x}{\sqrt{3}}$	**B1 B1**	**B1** Reflection. **B1** In $y = \frac{x}{\sqrt{3}}$; accept $y = \frac{\sqrt{3}}{3}x$, $y = x\tan 30°$, or the line through the origin at $30°$ or $\frac{\pi}{6}$ to the x-axis.
	Stretch parallel to the y-axis, scale factor 6	**B1**	Accept in the y-direction, or with the x-axis invariant.
	Reflection first, then the stretch	**B1**	Order stated explicitly, with both transformations named.
		[4]	
5(b)	Inverts each factor, in reverse order: $\mathbf{M}^{-1} = \begin{pmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix}^{-1} \begin{pmatrix} 1 & 0 \\ 0 & 6 \end{pmatrix}^{-1}$	**M1**	Multiplies the inverses of the two factors in reverse order; condone slips in either inverse. Not for the original order.
	$\mathbf{M}^{-1} = \begin{pmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & \frac{1}{6} \end{pmatrix}$	**A1**	Both inverses evaluated, in this order. A0 for a single multiplied-out matrix.
		[2]	
5(c)	$\mathbf{M} = \begin{pmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} \\ 3\sqrt{3} & -3 \end{pmatrix}$	**B1**	
	"$\mathbf{M}$" $\begin{pmatrix} 1 \\ m \end{pmatrix} = k \begin{pmatrix} 1 \\ m \end{pmatrix}$ $\Rightarrow \frac{1}{2} + \frac{\sqrt{3}}{2}m = k,\ 3\sqrt{3} - 3m = km$	**M1**	Applies their $\mathbf{M}$ to (x, mx) or $(1, m)$ and sets the image on $y = mx$, giving two equations.
	$3\sqrt{3} - 3m = m\left(\frac{1}{2} + \frac{\sqrt{3}}{2}m\right)$	**M1**	Eliminates k, or substitutes the image into $y = mx$, to form an equation in m alone.
	$\sqrt{3}m^2 + 7m - 6\sqrt{3} = 0$	**A1**	oe
	$(\sqrt{3}m - 2)(m + 3\sqrt{3}) = 0 \Rightarrow m = \ldots$	**M1**	Solves their three-term quadratic by factors whose constants multiply to their constant term, the formula with their values, or completing the square. Calculator allowed.
	$y = \frac{2}{\sqrt{3}}x, \quad y = -3\sqrt{3}x$	**A1 A1**	**A1** $y = \frac{2}{\sqrt{3}}x$ oe, or awrt $y = 1.15x$. An equation, not m alone. **A1** $y = -3\sqrt{3}x$ oe, or awrt $y = -5.20x$. An equation, not m alone.
		[7]	
	Total for Question 5	**13**	

Question	Scheme	Marks	Notes
6(a)	$\begin{pmatrix}2 & -1\\ p & 0\end{pmatrix}\begin{pmatrix}0\\ y\end{pmatrix}=\begin{pmatrix}-y\\ 0\end{pmatrix}$ not on $x=0$ for $y\neq 0$, so not invariant	**B1**	Image of a general point, or of a specific point other than the origin, with a statement that it is not on $x=0$.
	$\begin{pmatrix}2 & -1\\ p & 0\end{pmatrix}\begin{pmatrix}x\\ mx\end{pmatrix}=\begin{pmatrix}(2-m)x\\ px\end{pmatrix}$ $\Rightarrow px=m(2-m)x$	**M1**	Finds the image of a general point (x, mx) and substitutes it into $y=mx$. Condone sign slips in the image.
	$\Rightarrow m^2-2m+p=0$	**A1**	Correct substituted equation leading without error to the printed result; combined steps accepted.
	No invariant lines $\Leftrightarrow (-2)^2-4p<0$	**M1**	Sets the discriminant of the quadratic in m negative, or completes the square. Condone slips in the discriminant.
	$p>1$	**A1**	Strict inequality, oe in set notation.
		[5]	
6(b)	$\det \mathbf{M}=p=-3$	**B1**	
	$m^2-2m-3=0\Rightarrow m=\ldots$	**M1**	Substitutes their value of p into $m^2-2m+p=0$ and solves the quadratic by any means; a calculator is allowed.
	$y=3x$ and $y=-x$	**A1**	Both, as equations of lines. Lines not through the origin need not be considered.
		[3]	
	Total for Question 6	**8**	

Question	Scheme	Marks	Notes
7(a)	$(x, mx) \mapsto \big(x - 4mx,\ 2x - mx\big)$	**M1**	Applies **B** to a general point of the line, such as (x, mx) or $(1, m)$, to find its image. Condone one slip.
	$\dfrac{Y}{X} = \dfrac{(2-m)x}{(1-4m)x} \Rightarrow y = \dfrac{2-m}{1-4m}x$	**A1 AG**	Correct image coordinates leading without error to the printed equation, by their ratio or by eliminating x.
	$y = \frac{1}{4}x \mapsto x = 0$ $x = 0 \mapsto y = \frac{1}{4}x$	**B1**	Both images, oe such as $4y = x$ or the y-axis.
		[3]	
7(b)	$m = \dfrac{2-m}{1-4m}$	**M1**	Equates m to the image gradient from (a).
	$4m^2 - 2m + 2 = 0$	**A1**	oe, such as $2m^2 - m + 1 = 0$.
	$(-2)^2 - 4(4)(2) = \ldots$	**M1**	Evaluates the discriminant of their three-term quadratic, or completes the square, to test for real roots.
	$-28 < 0 \Rightarrow$ no real m $y = \frac{1}{4}x$ and $x = 0$ map to each other, so neither is invariant No invariant line through the origin, so none parallel to one: T has no invariant lines	**A1 AG**	Correct negative discriminant, both special lines shown not invariant from (a), and a conclusion citing the parallel-line fact. A0 if either special line is omitted.
		[4]	
Total for Question 7		**7**	

Question	Scheme	Marks	Notes
8(a)	$\mathbf{A}^2 = \begin{pmatrix} b^2+(a-3)(2a-1) & (a-3)(b-2) \\ (2a-1)(b-2) & (2a-1)(a-3)+4 \end{pmatrix}$	**M1**	Multiplies **A** by itself and sets the result equal to **I**, or equates **A** to its inverse by formula. Condone slips in entries.
	$b^2+(a-3)(2a-1)=1$, $(a-3)(b-2)=0$ $(2a-1)(b-2)=0$, $(2a-1)(a-3)+4=1$	**A1**	All four equations correct; need not be simplified.
	$2a^2-7a+6=0 \Rightarrow a=\ldots$	**dM1**	Forms a three-term quadratic in a from the bottom-right entry and solves it by any means.
	$a=\frac{3}{2}$ or $a=2$	**A1**	Both values, and no others.
	$a\neq 3 \Rightarrow (a-3)(b-2)=0 \Rightarrow b=2$	**A1**	$b=2$ only, from an off-diagonal equation, with $a-3$ or $2a-1$ shown or stated non-zero for both values of a.
		[5]	
8(b)	$\begin{pmatrix} 2 & -\frac{3}{2} \\ 2 & -2 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix}=\begin{pmatrix} x \\ y \end{pmatrix}$	**M1**	Uses **A** with their smaller a and their b from (a), sets the image of (x,y) equal to (x,y) and forms two equations.
	$2x-\frac{3}{2}y=x \Rightarrow 2x-3y=0$ $2x-2y=y \Rightarrow 2x-3y=0$ Same equation, so a line: $2x-3y=0$	**A1**	Both equations reduced to the same line and stated equivalent, giving $2x-3y=0$ oe, such as $y=\frac{2}{3}x$.
		[2]	
	Total for Question 8	**7**	

Question	Scheme	Marks	Notes
9(i)	$\begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 2 \\ 0 \end{pmatrix}$	**M1**	Equates the image of $(1, -1)$ under a general matrix to $(2, 0)$, forming equations in the entries. Condone one slip.
	$a - b = 2, \quad c - d = 0$	**A1**	Both, oe.
	$\lvert\det \mathbf{A}\rvert = \frac{24}{4} \Rightarrow \lvert ad - bc\rvert = 6$	**M1**	Equates $\det \mathbf{A}$ or $\lvert\det \mathbf{A}\rvert$ to $\frac{24}{4}$; $\det \mathbf{A} = 6$ alone, or -6 alone, is enough.
	$(b + 2)d - bd = \pm 6 \Rightarrow 2d = \pm 6$	**A1**	Correct equation in the remaining entries, oe such as $2c = \pm 6$; one sign alone accepted.
	$\det(\mathbf{A} - \mathbf{I}) = (a - 1)(d - 1) - bc = 0$	**M1**	Uses $\det(\mathbf{A} - \mathbf{I}) = 0$, or another valid condition for a line of invariant points, with their entries. Condone sign slips.
	$d - b - 1 = 0$	**A1**	oe, in any entries.
	$d = 3 \Rightarrow b = \ldots, \; a = \ldots, \; c = \ldots$	**dM1**	Solves their three equations together, choosing a sign where needed, to find all four entries.
	$\mathbf{A} = \begin{pmatrix} 4 & 2 \\ 3 & 3 \end{pmatrix}$	**A1**	Or $\begin{pmatrix} -2 & -4 \\ -3 & -3 \end{pmatrix}$.
		[8]	
9(ii)	$\begin{pmatrix} 2 & 1 \\ 2 & 3 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} \Rightarrow 2x + y = x$	**M1**	Sets the image of (x, y) under $\mathbf{B}$ equal to (x, y) and forms at least one equation. Condone one slip.
	$y = -x$	**A1**	oe such as $x + y = 0$.
		[2]	
9(iii)	$\begin{pmatrix} 2 & 1 \\ 2 & 3 \end{pmatrix}\begin{pmatrix} x \\ 2x + c \end{pmatrix} = \ldots$	**M1**	Multiplies $\mathbf{B}$ by a general point of the line, such as $(t, 2t + c)$. Condone one slip.
	$= \begin{pmatrix} 4x + c \\ 8x + 3c \end{pmatrix}$	**A1**	
	$2(4x + c) + c = 8x + 3c$ so the image lies on $y = 2x + c$: invariant	**A1 AG**	Correct substitution showing the image satisfies $y = 2x + c$, without error, and a conclusion that the line is invariant.
		[3]	
	Total for Question 9	**13**	